Research Report: Quantum mechanics

1. Executive Summary

Quantum mechanics, also known as quantum physics, is the fundamental physical theory that describes the behavior of matter and of light; the behaviors it models typically occur at and below the scale of atoms, and have often been described as counterintuitive. Its concepts and methods have been applied across many disciplines, including quantum chemistry, quantum biology, quantum field theory, quantum technology, and quantum information science.

2. Related Key Insights

- Quantum mechanics Category

- Einstein's thought experiments - A hallmark of Albert Einstein's career was his use of visualized thought experiments as a fundamental tool for understanding physical issues and for elucidating his concepts to others. Einstein's thought experiments took diverse forms. In his youth, he mentally chased beams of light.

- Macroscopic quantum phenomena - Macroscopic quantum phenomena are processes showing quantum behaviour at the macroscopic scale, rather than at the atomic scale where quantum effects are prevalent.